Exercise Sheet #1: Basic Concepts

Discussion: Oct 15, 2025

Exercise 1: Coulomb Force

Two point particles are separated by a distance of 0.60 m and carry a total charge of 200 μC. Determine the charge of each of the two particles if they

a) repel each other with a force of 80 N,

b) attract each other with a force of 80 N.

Solution

Let the charges be q_1 and q_2 with

$$q_1 + q_2 = Q = 200 \cdot 10^{-6}\,\mathrm{C}.$$

The Coulomb force is

$$\begin{aligned} F &= k\frac{|q_1 q_2|}{r^2}, \\ k &= 8.99 \cdot 10^9\,\mathrm{N\,m^2/C^2}, \\ r &= 0.60\,\mathrm{m}, \\ F &= 80\,\mathrm{N}. \end{aligned}$$

Thus,

$$|q_1 q_2| = \frac{Fr^2}{k} \approx 3.2 \cdot 10^{-9}\,\mathrm{C}^2,$$

with

$$q_1 q_2 = q_1(q - q_1).$$

a)

$$q_1 q_2 = +3.2 \cdot 10^{-9} \Rightarrow x^2 - (200 \cdot 10^{-6})x + 3.2 \cdot 10^{-9} = 0$$

With

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

$$q_1 \approx 1.8 \cdot 10^{-4}\,\mathrm{C}, \quad q_2 \approx 1.8 \cdot 10^{-5}\,\mathrm{C}.$$

b)

$$q_1 q_2 = -3.2 \cdot 10^{-9} \Rightarrow x^2 - (200 \cdot 10^{-6})x - 3.2 \cdot 10^{-9} = 0$$

$$q_1 \approx -1.5 \cdot 10^{-5}\,\mathrm{C}, \quad q_2 \approx 2.1 \cdot 10^{-4}\,\mathrm{C}.$$

Reversed signs are also possible.

Exercise 2: Ohm's Law

A resistor of $15\,\Omega$ is connected to a voltage of $30\,\mathrm{V}$. What are the current and the conductance?

Solution

Ohm's law states:

$$I = \frac{U}{R}$$

Substituting the given values:

$$I = \frac{30\,\mathrm{V}}{15\,\Omega} = 2\,\mathrm{A}$$

The conductance is the reciprocal of the resistance:

$$G = \frac{1}{R} = \frac{1}{15\,\Omega} \approx 0.067\,\mathrm{S}$$

Exercise 3: Resistance

Given are two copper wires of equal mass with circular cross-sections; wire A is twice as long as wire B. How do the resistances of the wires compare (neglecting temperature effects)?

a) $R_A = 8R_B$,

b) $R_A = 4R_B$

c) $R_A = 2R_B$,

d) $R_A = R_B$.

Solution

The resistance of a wire is given by

$$R = \rho \frac{l}{A},$$

where ρ is the resistivity, l is the length, and A is the cross-sectional area.
Since both wires have the same mass and material, we have

$$m = \rho_m V = \rho_m (A \cdot l) = \text{const.}$$

Thus, $A \cdot l = \text{const.}$, which implies $A \propto \frac{1}{l}$.
For wire A, the length is $l_A = 2l_B$. Therefore, its cross-sectional area is

$$A_A = \frac{1}{2} A_B.$$

Now the resistance ratio becomes

$$\frac{R_A}{R_B} = \frac{\rho \frac{l_A}{A_A}}{\rho \frac{l_B}{A_B}} = \frac{l_A}{l_B} \cdot \frac{A_B}{A_A} = \frac{2l_B}{l_B} \cdot \frac{A_B}{\frac{1}{2} A_B} = 2 \cdot 2 = 4.$$

Therefore,

$$R_A = 4R_B.$$

Exercise 4: Resistivity

Suppose we have a copper wire with a resistance of $2\,\Omega$. We want to replace the copper wire with an aluminum wire of the same length and resistance. By what factor must the diameter of the wire be increased?

Solution

For a cylindrical wire,

$$A = \frac{\pi d^2}{4},$$

where d is the diameter.
We require the aluminum wire to have the same resistance R as the copper wire, and both wires have the same length l. Thus,

$$\rho_{\mathrm{Cu}} \frac{l}{A_{\mathrm{Cu}}} = \rho_{\mathrm{Al}} \frac{l}{A_{\mathrm{Al}}}.$$

Canceling l,

$$\frac{A_{\mathrm{Al}}}{A_{\mathrm{Cu}}} = \frac{\rho_{\mathrm{Al}}}{\rho_{\mathrm{Cu}}}.$$

Since $A \propto d^2$,

$$\left(\frac{d_{\mathrm{Al}}}{d_{\mathrm{Cu}}}\right)^2 = \frac{\rho_{\mathrm{Al}}}{\rho_{\mathrm{Cu}}}.$$

Therefore,

$$\frac{d_{\mathrm{Al}}}{d_{\mathrm{Cu}}} = \sqrt{\frac{\rho_{\mathrm{Al}}}{\rho_{\mathrm{Cu}}}}.$$

Numerical values of resistivity:

$$\rho_{\mathrm{Cu}} \approx 0.017\,\frac{\Omega \cdot \mathrm{mm}^2}{\mathrm{m}}, \quad \rho_{\mathrm{Al}} \approx 0.027\,\frac{\Omega \cdot \mathrm{mm}^2}{\mathrm{m}}.$$

Thus,

$$\frac{d_{\mathrm{Al}}}{d_{\mathrm{Cu}}} = \sqrt{\frac{0.027}{0.017}} \approx 1.26.$$

$$\boxed{\text{The diameter of the aluminum wire must be about 1.3 times larger.}}$$

Exercise 5: Temperature Dependence of Resistance

At what temperature does a copper wire have a resistance that is 10% greater than at 20 °C?

Solution

The temperature dependence of the resistance is given by

$$R(T) = R_0\big(1 + \alpha(T - T_0)\big),$$

where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.
We are looking for the temperature T at which

$$R(T) = 1.1\,R_0.$$

Thus,

$$1.1R_0 = R_0\big(1 + \alpha(T - 20°\mathrm{C})\big).$$

Dividing through by R_0:

$$1.1 = 1 + \alpha(T - 20).$$

Hence,

$$T - 20 = \frac{0.1}{\alpha}.$$

For copper, $\alpha \approx 0.0039\,\mathrm{K}^{-1}$, so

$$T \approx 20 + \frac{0.1}{0.0039} \approx 45.6°\mathrm{C}.$$

Therefore, the copper wire has a 10% greater resistance at approximately

$$\boxed{46°\mathrm{C}}$$

than at 20 °C.

Exercise 6: Energy

A rechargeable flashlight battery is capable of delivering 90 mA for about 12 h. How much charge can it release at that rate? If its terminal voltage is 1.5 V, how much energy can the battery deliver?

Solution

The battery delivers a current of

$$I = 0.09\,\mathrm{A},$$

for a time of

$$t = 12 \cdot 3600\,\mathrm{s} = 43200\,\mathrm{s}.$$

a)
$$Q = I \cdot t = 0.09 \times 43200 = 3888\,\mathrm{C}.$$

So the total charge is

$$Q \approx 3.9\,\mathrm{kC}.$$

b) The energy is given by

$$W = Q \cdot V,$$

where $V = 1.5\,\mathrm{V}$. Thus,

$$W = 3888 \times 1.5 = 5832\,\mathrm{J}.$$

So the total energy is

$$W \approx 5.8\,\mathrm{kJ}.$$

Exercise 7: Power

Find the hot resistance of a light bulb rated $60\,\mathrm{W}$, $120\,\mathrm{V}$.

Solution

The electrical power is related to the voltage and resistance by

$$P = \frac{V^2}{R}.$$

Rearranging for R,

$$R = \frac{V^2}{P}.$$

Substitute the given values:

$$V = 120\,\mathrm{V}, \quad P = 60\,\mathrm{W}.$$

$$R = \frac{(120)^2}{60} = \frac{14400}{60} = 240\,\Omega.$$

Thus, the *hot resistance* of the bulb is $240\,\Omega$.

Remark: The filament of a light bulb has a much lower resistance when it is cold, typically about

$$R_{\text{cold}} \approx \frac{R_{\text{hot}}}{10} \sim 20\text{–}30\,\Omega,$$

depending on the material and temperature coefficient of resistance. This explains the large inrush current when the bulb is first switched on.

Exercise 8: Particles

A particle accelerator produces a proton beam with a current of $3.50\,\mu\mathrm{A}$. Each proton has an energy of $60.0\,\mathrm{MeV}$. The protons strike a target in a vacuum chamber and come to rest in it, heating the target.

a) How many protons strike the target per second?

b) How much energy is deposited in the target per second?

Solution

The beam current is $I = 3.50\,\mu\text{A} = 3.50 \times 10^{-6}\,\text{A}$. The elementary charge is $e = 1.602 \times 10^{-19}\,\text{C}$.

a) The number of protons per second is

$$\dot{N} = \frac{I}{e}.$$

Substituting,

$$\dot{N} = \frac{3.50 \times 10^{-6}}{1.602176634 \times 10^{-19}} \approx 2.18 \times 10^{13}\,\text{s}^{-1}.$$

b) Each proton has an energy of $60.0\,\text{MeV}$.

The deposited energy per second is

$$P = V \cdot I = \frac{W}{Q} \cdot I = \frac{60.0\,\text{MeV}}{e} \cdot 3.50\,\mu\text{A} = 210\,\text{W}.$$